STUDENT PERFORMANCE ANALYTICS

MINI PROJECT REPORT

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GitHub Repository
github.com/vanshkohli225/student-performance-analytics

1. Project Overview

Student Performance Analytics is a Python-based data analysis project that examines students' marks across Maths, Science and English. It calculates performance measures, assigns grades, checks pass/fail status, identifies top performers and presents the results through charts.

2. Objectives

- Calculate each student's total marks, average and percentage.
- Assign grades using percentage-based grade ranges.
- Determine pass/fail status using a minimum score of 40 in every subject.
- Find class-level statistics and identify the top three performers.
- Compare average marks across subjects and visualize the results.

3. Technologies Used

Technology	Purpose
Python	Core programming and project logic
NumPy	Array operations and numerical calculations
Pandas	Dataset loading, tabular processing and report export
Matplotlib	Charts for subject averages and student totals

Technology	Purpose
Google Colab	Cloud-based notebook environment for running the project

4. Dataset Description

The notebook uses a sample dataset containing 20 student records. Each record includes a student ID, student name and marks in three subjects: Maths, Science and English. Each subject is scored out of 100. The sample data is included in the notebook so the project can be executed without uploading an external file.

5. Methodology

Data loading: Create a Pandas DataFrame from the sample student marks and select the three subject columns.

Numerical analysis: Convert subject marks to a NumPy array and calculate totals, averages and percentages.

Grade assignment: Apply a function that assigns A+, A, B, C, D or F based on percentage ranges.

Result checking: Use NumPy conditions to mark a student Pass only when every subject score is at least 40.

Performance analysis: Calculate class statistics, identify highest and lowest totals, and list the top three students.

Visualization and export: Plot subject averages and student totals, then save the processed table as a CSV report.

6. NumPy and Pandas Functions Used

Function / Method	Use in the project
to_numpy()	Converts selected Pandas columns into a NumPy array.
np.sum()	Calculates total marks for each student.
np.mean()	Calculates averages.
np.max() / np.min()	Finds the highest and lowest marks.
np.median()	Finds the median mark.
np.std()	Measures spread of marks using standard deviation.
np.where()	Assigns Pass/Fail based on subject-wise conditions.
flatten()	Combines marks into a one-dimensional array for class statistics.
nlargest()	Returns the three students with the highest total marks.
to_csv()	Exports the processed analysis to a CSV file.

7. Sample Results

The following results are calculated from the 20 sample student records included in the Colab notebook.

Metric	Result
Number of students	20
Class average across all subject marks	78.28
Highest individual subject mark	97
Lowest individual subject mark	49
Students passed	20
Students failed	0

Top 3 performers (by total marks)

Student	Total / 300	Percentage	Grade
Pari	285	95.00%	A+
Meera	278	92.67%	A+
Diya	275	91.67%	A+

8. Visualizations

The notebook generates two Matplotlib charts: (1) subject-wise average marks, which compares the class average in Maths, Science and English; and (2) total marks by student, which compares overall performance and makes the highest-scoring students easy to identify.

9. Output File

After execution, the notebook exports the enriched dataset as **student_performance_report.csv**. The CSV includes the original student details along with Total, Average, Percentage, Grade and Result columns.

10. Conclusion

This project demonstrates a basic data analysis workflow using Python libraries. NumPy is used for numerical operations, Pandas for handling and transforming tabular data, and Matplotlib for visual presentation. The analysis summarizes student performance and highlights overall and subject-wise trends.

11. GitHub Repository

https://github.com/vanshkohli225/student-performance-analytics

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